# Brian Benegas

Full Stack Developer
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## SUMMARY

Full stack developer building and operating public-sector systems that run unattended all day and cannot go down. I work end to end: backend and frontend, plus the Linux hosts, CI/CD pipelines, nginx and certificates that put them in production. Most of my work is hardening real-time systems under load — horizontal scaling, memory behaviour in long-running clients, and database access at peak. Currently leading the innovation team at the Municipality of Ituzaingó.

## EXPERIENCE

### Municipality of Ituzaingó — Secretariat of Innovation and Technology

Buenos Aires, Argentina · September 2023 – Present
**Innovation Team Lead** (October 2024 – Present) · **Backend Developer** (September 2023 – October 2024)

Primary engineer on VUS, a real-time queue management system running daily across municipal offices: self-service kiosks, thermal ticket printing, waiting-room displays, operator stations, a live dashboard and historical analytics, deployed on Kubernetes.

- Resolved intermittent real-time event loss when scaling to multiple Kubernetes replicas by introducing Redis as a Socket.IO pub/sub adapter, removing per-process in-memory connection state that silently made pods non-interchangeable.
- Eliminated daily peak API timeouts by rewriting a non-SARGable query (DATE(column) = CURDATE()) as a pure range scan, restoring index usage and reducing a full table scan to milliseconds.
- Flattened memory growth on always-on waiting-room displays by replacing an unbounded Set with a 60-second LRU Map, and removed hand-rolled reconnection watchdogs in favour of Socket.IO's native exponential backoff.
- Eliminated duplicate call announcements with an ID-plus-state hash debounce over a 15-second anti-repeat window.
- Decoupled browsers from operating-system printer ports by building a local .NET daemon that translates JSON over HTTP into native ESC/POS commands, isolating thermal printer drivers behind a single replaceable component.
- Split a false monorepo — four microservices sharing one .git — into independent repositories with their own GitLab CI/CD pipelines, ending dependency collisions and cross-triggered builds.
- Separated a socket-driven live operations dashboard from historical analytics, so heavy aggregate queries never run on the real-time path.
- Made offices, services and queue behaviour runtime-configurable, so per-institution changes ship as data rather than as a new build.
- Built a recurring Node.js ETL migrating a monthly social-benefit roster from Excel into MySQL, keeping personal-data files out of version control by design and requiring explicit file selection through an interactive CLI to prevent silent reprocessing.
- Lead the innovation team, coordinate delivery between engineering and administrative departments, and train public officials on the systems shipped.

### Pidgeon Solutions — Co-founder

- Co-founded a software services studio, working with a second engineer and a graphic designer.

## SELECTED PROJECTS

**VUS — Municipal queue management system**
NestJS · SvelteKit · TypeScript · MySQL · TypeORM · Socket.IO · Redis · Kubernetes · .NET · GitLab CI/CD
Full ecosystem for public-facing service desks: touch kiosk, orchestration API, waiting-room screens, operator panel, ESC/POS print daemon, live dashboard and analytics. Built for heterogeneous hardware and unattended operation, where every component has to degrade without taking the rest down.

**brianleft.com — Interactive portfolio with a hand-written RAG pipeline**
SvelteKit · NestJS · TypeScript · MySQL · Redis · Docker · nginx
Portfolio navigated as a Linux console, answering questions about my work through a retrieval pipeline written step by step in TypeScript — document ingestion, LLM keyword extraction, relevance retrieval, context assembly and streaming — with no workflow engine in between. Self-hosted end to end on my own VPS, including nginx, TLS and deployment.

**SØREN — Computer vision over an unattended process**
Python · OpenCV · Tesseract · Discord Webhooks
Reads an external process's state through OCR and pixel analysis, without touching its memory or network traffic, and acts on it. Runs for hours unsupervised; the interesting part is not the automation but recovering on its own when conditions change underneath.

**Søren Mirror — Reading a chess board from screen pixels**
Python · OpenCV · MSS · python-chess · Stockfish
Desktop overlay that reconstructs game state by computer vision rather than DOM scraping or code injection, deducing board orientation, colour and turn geometrically, then analysing the position with Stockfish in a floating HUD.

## SKILLS

**Languages** · TypeScript, JavaScript, C#, Python, SQL
**Backend** · Node.js, NestJS, .NET, TypeORM, Socket.IO, REST APIs
**Frontend** · SvelteKit, Svelte
**Data** · MySQL, MongoDB, Redis
**Infrastructure** · Docker, Kubernetes, nginx, Linux (Debian, Ubuntu, Arch), GitLab CI/CD, PM2, VPS administration
**Practices** · SOLID, clean code, defensive programming for 24/7 systems, Git, Jira, Postman

## EDUCATION

**Technical Degree in Software Development** — Regional University Center of Ituzaingó · 2023 – Present

## LANGUAGES

Spanish (native) · English (advanced)

## COURSES

Platzi, 2024 — C# programming, .NET fundamentals, data management with LINQ, SOLID principles, object-oriented programming, best practices and clean code, advanced TypeScript types and functions.